Check for updates

Royal Australasian College of Surgeons

Using artificial intelligence to predict choledocholithiasis: can machine learning models abate the use of MRCP in patients with biliary dysfunction?

Joshua Blum,* Sam Hunn,* Jules Smith,* Fa Yu Chan† and Richard Turner*†

*Department of General Surgery, Royal Hobart Hospital, Hobart, Tasmania, Australia and
†Tasmanian School of Medicine, University of Tasmania, Hobart, Tasmania, Australia

Key words
artificial intelligence, biliary tract diseases, general surgery, machine learning, prediction.

Correspondence
Dr Joshua Blum, Department of General Surgery, Royal Hobart Hospital, Hobart, Tasmania, Australia.
Email: joshua.blum@ths.tas.gov.au

J. Blum MBBS; **S. Hunn** MBBS; **J. Smith** MBBS; **F. Y. Chan** MBBS; **R. Turner** MBBS (Hons), BMedSc, FRACS, PhD, Grad Cert Data Science.

Accepted for publication 7 March 2024.

doi: 10.1111/ans.18950

Abstract

Background: Prompt diagnosis of choledocholithiasis is crucial for reducing disease severity, preventing complications and minimizing length of stay. Magnetic resonance cholangiopancreatography (MRCP) is commonly used to evaluate patients with suspected choledocholithiasis but is expensive and may delay definitive intervention. To optimize patient care and resource utilization, we have developed five machine learning models that predict a patients' risk of choledocholithiasis based on clinical presentation and pre-MRCP investigation results.

Methods: Inpatients admitted to the Royal Hobart Hospital from 2018 to 2023 with a suspicion of choledocholithiasis were included. Exclusion criteria included prior hepatobiliary surgery, known hepatobiliary disease, or incomplete records. Variables related to clinical presentation, laboratory testing, and sonographic or CT imaging were collected. Four machine learning techniques were employed: logistic regression, XGBoost, random forest, and K-nearest neighbours. The three best performing models were combined to create an ensemble model. Model performance was compared against the American Society for Gastrointestinal Endoscopy (ASGE) choledocholithiasis risk stratification guidelines.

Results: Of the 222 patients included, 113 (50.9%) had choledocholithiasis. The most successful models were the random forest (accuracy: 0.79, AUROC: 0.83) and ensemble (accuracy and AUROC: 0.81). Every model outperformed the ASGE guidelines. Key variables influencing the models' predictions included common bile duct diameter, lipase, imaging evidence of cholelithiasis, and liver function tests.

Conclusion: Machine learning models can accurately assess a patient's risk of choledocholithiasis and could assist in identifying patients who could forgo an MRCP and proceed directly to intervention. Ongoing validation on prospective data is necessary to refine their accuracy and clinical utility.

Introduction

Prompt diagnosis of choledocholithiasis in patients who present with acute biliary dysfunction is vital in ensuring maximal reduction in the risk of complication and disease severity. In the absence of computed topography (CT) or ultrasound scanning (USS) findings conclusive for choledocholithiasis, patients presenting with acutely cholestatic liver enzymes and/or acute obstructive jaundice often undergo magnetic resonance cholangiopancreatography (MRCP) to further evaluate their presentation.[1] MRI scanning is expensive and often limited to larger facilities.[2] Patients with choledocholithiasis who receive an MRCP prior to endoscopic retrograde cholangiopancreatography (ERCP) experience prolonged admissions, contribute to increased hospital expenses, and experience greater rates of in-hospital complication.[3] This highlights the importance of timely identification of patients who possess a high risk of choledocholithiasis, as early intervention may contribute to improved patient and financial outcomes.

The American Society for Gastrointestinal Endoscopy (ASGE) guidelines posit that the finding of a dilated common bile duct (CBD) on imaging in conjunction with a bilirubin greater than 68.4 mmol/L deems a patient as being high risk of having

© 2024 Royal Australasian College of Surgeons.

choledocholithiasis, for which they recommended patients proceed directly to ERCP.[4] Adherence to these guidelines has been associated with reduced hospital expenditure.[5] Despite this, it is apparent that a large proportion of high-risk patients receive additional unnecessary imaging prior to ERCP.[6]

To address this disparity between guidelines and clinical practice, we have developed a series of predictive models that use machine learning techniques which aim to accurately identify patients—based on pre-MRCP clinical, biochemical, and imaging results—who are at highest risk of choledocholithiasis and therefore would most likely benefit from timely intervention without needing an MRCP.

Methods

Ethics approval

Ethics approval for this project was granted by the University of Tasmania Human Research Ethics Committee (project ID 28762).

Study design

A retrospective cohort analysis was conducted on inpatients admitted between 2018 and 2023 with a clinical suspicion of acute choledocholithiasis and who received either an MRCP, ERCP or both during their admission. Patients were excluded if they had prior hepatobiliary surgery, known chronic hepatobiliary disease—including malignancy—prior to their acute admission, or incomplete medical records.

Data and included variables

Data were collected from patients' digital medical records and are listed in Table 1. Demographic variables were age and sex. Variables regarding clinical presentation were the presence of right-upper quadrant/epigastric pain, subjective fevers, jaundice, and vital signs on presentation. The values of relevant laboratory investigations upon initial presentation to hospital were noted, namely white cell count (WCC), neutrophils, C-reactive protein (CRP), alkaline phosphatase (ALP), aspartate transaminase (AST), alanine transaminase (ALT), gamma-glutamyl transferase (GGT), lipase, bilirubin, and blood cultures. Imaging results collected were CT or USS report documentation of cholelithiasis, choledocholithiasis, cholecystitis, common bile duct (CBD) diameter, and evidence of a potential malignancy. Where CBD diameter was absent from the report, this information was collected directly from the images. A patient's final diagnosis was sourced from the report of their definitive intervention; in cases where no operative intervention was performed, the diagnosis was ascertained from their discharge summary.

Machine learning analysis

The collected data were pre-processed to develop four distinct datasets, demonstrated in Table 1. Dataset A consisted of continuous variables, standardized using the Z-score method, as well as binary variables. In Dataset B, continuous variables were converted to binary variables based on whether they exceeded the relevant standard laboratory cut-off values. A cut-off of 55 years was selected for age. Binary variables in Dataset B that were of statistical significance between patients with and without choledocholithiasis were identified via Chi-Squared testing and used to create Dataset C. Continuous variables that were significantly different between patients with and without choledocholithiasis were identified via two-way t-testing, and were standardized and used in conjunction with significant binary variables to form Dataset D. Bilirubin was excluded from the binary dataset as all patients exceeded the cut-off value.

Table 1 Variables collected

Continuous variables	Binary variables	
Age[A,B]	Age >55[B]	Objectively febrile[A,B,C,D]
Systolic blood pressure[A]	Sex[A,B,C,D]	ALP >130[B,C]
Diastolic blood pressure[A,D]	Right upper quadrant pain[A,B]	ALT >45[B,C]
Heart rate[A]	Subjective fever[A,B]	AST >40[B,C]
Respiratory rate[A]	Jaundice[A,B]	GGT >60[B,C]
ALP[A,D]	Hypertensive[B]	CRP >5
ALT[A,D]	Hypotensive[B,C]	Lipase >180[B,C]
AST[A,D]	Diastolic pressure high[B]	WCC >11
GGT[A,D]	Diastolic pressure low[B]	Neutrophils >7.5
Lipase[A,D]	Tachycardic[B]	Cholelithiasis on USS/CT[A,B,C,D]
Peak bilirubin[A]	Bradycardic[B]	Choledocholithiasis on US/CT[A,B,C,D]
CRP[A]	Tachypnoeic[B]	CBD Diameter >6mm[B,C]
White cell count (WCC)[A]	Bradypnoeic[B]	Malignancy on USS/CT[A,B,C,D]
Neutrophil count[A]		
CBD Diameter on USS/CT[A,D]		

Dataset A: *A*, Dataset B: *B*, Dataset C: *C*, Dataset D: *D*.

Four machine learning techniques were employed: logistic regression (LR), k-nearest neighbours (KNN), random forest (RF), and XGBoost. An ensemble model combining the outputs of the LR, KNN, and RF models was created by combining each via a 'vote'; if two or more models predicted a case as being positive, the final ensemble prediction would also be positive; otherwise, it would be negative.

The collected data was split into training and testing sets, with a 70/30 split. The randomization of patients to either training or testing data occurred identically across all four datasets. Hyperparameter tuning was conducted to identify the optimal value of 'K' for the KNN model, as well as the optimal number of variables to sample at each split for the RF model.

Model performance was evaluated via receiver-operator characteristic (ROC) curves, area-under-ROC (AUROC), accuracy, specificity, sensitivity, negative predictive value (NPV), positive predictive value (PPV) and F1 score. The performance of these models was compared to the performance of the ASGE choledocholithiasis risk stratification guidelines' 'high risk' category for probability of choledocholithiasis.[4]

The mean decrease in Gini coefficient was calculated to identify which variables were the most important predictors of choledocholithiasis for the RF models. Statistically significant ($p < 0.05$) logit coefficients were recorded from the LR models.

© 2024 Royal Australasian College of Surgeons.

14452197, 2024, 7-8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ans.18950 by University Of Southampton, Wiley Online Library on [18/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Results

Of the 222 patients eligible for analysis, 120 were male (54%). The mean (SD) age was 63.6 (18.7). 113 (50.9%) patients were proven to have choledocholithiasis as their primary diagnosis.

The RF model performed most consistently across the four datasets, with an AUROC of 0.83 for datasets A and B, and 0.77 and 0.76 for datasets C and D, respectively (Table 2, Fig. 1). The ensemble model utilizing dataset B had a peak accuracy of 0.81 with an AUROC of 0.81 (Fig. 1). All models were more sensitive than they were specific, with both the RF and the ensemble model achieving a peak sensitivity of 0.94 for dataset B (Table 2).

All models across each dataset performed more accurately than the high-risk stratification of the ASGE guidelines. In particular, the models were more specific whilst maintaining a comparable or better sensitivity (Table 2).

Lipase and common bile duct (CBD) diameter were consistently important variables for the RF model (Fig. 2). GGT and ALT were important in datasets featuring continuous variables but dropped in importance for binary datasets (Fig. 2). Imaging evidence of choledocholithiasis and cholelithiasis as well as AST were important for binary datasets but reduced in datasets featuring continuous variables (Fig. 2). Variables relating to clinical presentation as well as the presence of malignancy on imaging were consistently unimportant (Fig. 2).

The LR models' statistically significant logit coefficients demonstrated that CBD diameter and the presence of cholelithiasis on imaging were consistently positively correlated with choledocholithiasis (Fig. 3). Lipase was consistently negatively correlated with choledocholithiasis (Fig. 3). AST was negatively correlated with choledocholithiasis in Dataset A, but positively correlated in both binary datasets (Fig. 3). The presence of an objective fever was positively correlated with choledocholithiasis in both continuous datasets, but its relationship was statistically insignificant in the binary datasets (Fig. 3).

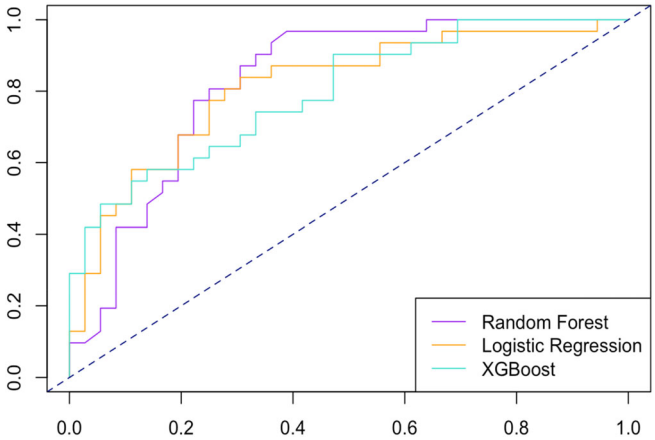

Fig. 1. Receiver operator curve for the random forest model in the prediction of choledocholithiasis when applied to Dataset B.

Discussion

The ability of these machine learning models to accurately predict the presence of choledocholithiasis based on clinical presentation and pre-MRCP investigation findings has many implications for clinical decision making. All models outperformed the current ASGE risk stratification guidelines across most of the datasets, suggesting that machine learning models may be a viable option when developing choledocholithiasis risk stratification tools. Such a tool may help clinicians accurately identify which patients would

Table 2 ASGE guidelines' performance at predicting choledocholithiasis

	AUROC	Sensitivity	Specificity	PPV	NPV	F1	Accuracy
	N/A	0.88	0.39	0.6	0.75	0.71	0.64
Model performance characteristics – Dataset A							
LR	0.75	0.66	0.72	0.72	0.66	0.69	0.69
XGBoost	0.76	0.71	0.69	0.71	0.69	0.71	0.7
RF	0.83	0.86	0.72	0.77	0.82	0.81	0.79
KNN	0.76	0.89	0.63	0.72	0.83	0.79	0.76
Ensemble	0.76	0.83	0.69	0.74	0.79	0.78	0.76
Model performance characteristics – Dataset B							
LR	0.81	0.81	0.69	0.69	0.81	0.75	0.75
XGBoost	0.8	0.71	0.67	0.65	0.73	0.68	0.69
RF	0.83	0.94	0.64	0.69	0.92	0.79	0.78
KNN	0.77	0.84	0.69	0.7	0.84	0.76	0.76
Ensemble	0.81	0.94	0.69	0.73	0.93	0.82	0.81
Model performance characteristics – Dataset C							
LR	0.76	0.8	0.79	0.74	0.76	0.77	0.75
XGBoost	0.65	0.66	0.6	0.64	0.61	0.65	0.63
RF	0.77	0.8	0.79	0.74	0.76	0.77	0.75
KNN	0.74	0.83	0.66	0.73	0.78	0.77	0.75
Ensemble	0.74	0.8	0.79	0.74	0.76	0.77	0.75
Model performance characteristics – Dataset D							
LR	0.76	0.83	0.66	0.74	0.78	0.77	0.75
XGBoost	0.76	0.77	0.66	0.71	0.72	0.74	0.72
RF	0.76	0.83	0.69	0.74	0.79	0.78	0.76
KNN	0.79	0.83	0.75	0.78	0.8	0.81	0.79
Ensemble	0.79	0.83	0.75	0.78	0.8	0.81	0.79

© 2024 Royal Australasian College of Surgeons.

14452197, 2024, 7-8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ans.18950 by University Of Southampton, Wiley Online Library on [18/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

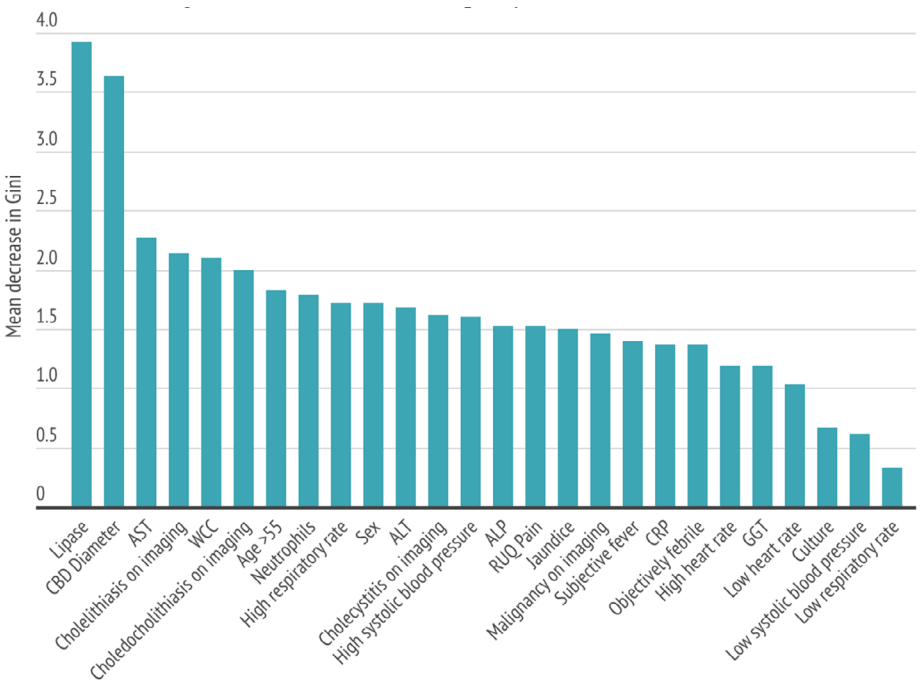

Fig. 2. Bar chart demonstrating mean decrease in Gini impurity for each variable in Dataset B.

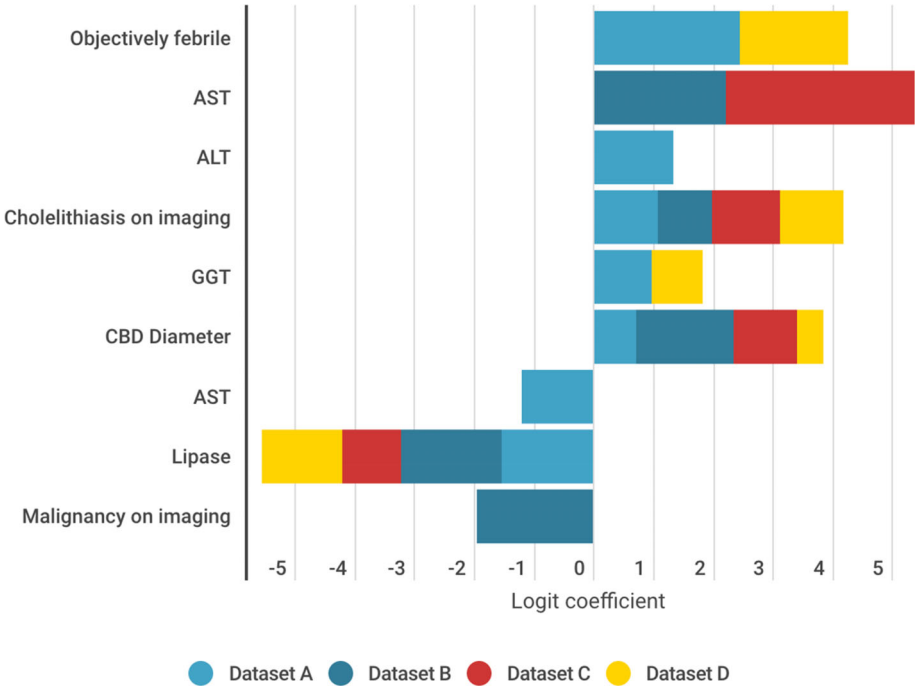

Fig. 3. Stacked bar chart demonstrating the statistically significant logit coefficients generated by the logistic regression model for each dataset.

benefit from proceeding directly to ERCP for definitive management of choledocholithiasis, thus reducing the reliance on MRCP in the diagnostic work-up of patients presenting with acute biliary dysfunction. Accurate identification of patients who are less likely to harbour choledocholithiasis would serve to prevent unnecessary ERCP as well as help identify patients who might benefit from further evaluation via MRCP.

The role of MRCP and CT

Our models, although powerful and inexpensive, have not yet surpassed the sensitivity and specificity of MRCP in the detection of choledocholithiasis.[7] As such, it is maintained that MRCP should be sought in cases where there is equivocal evidence for the presence of choledocholithiasis, as to mitigate the burden of ERCP-related complications and prolonged admission.[4] However, unpublished local

© 2024 Royal Australasian College of Surgeons.

14452197, 2024, 7-8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ans.18950 by University Of Southampton, Wiley Online Library on [18/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

audit data reveals that patients undergoing MRCP before ERCP for choledocholithiasis have longer admissions by 4 days and a higher rate of non-ERCP-related complications (36% *vs*. 1%) compared to those undergoing ERCP directly.[8] This increase in complications could be due to extended admission as well as the presence of co-morbidities in the former group that necessitates a cautious approach to management that minimizes the use of unnecessary ERCP. This highlights the increased hospital costs and poor outcomes associated with reduced MRCP accessibility, thus underscoring the importance of cheap, accessible alternatives to MRCP such as machine learning models.

CT scanning is more sensitive but less specific for choledocholithiasis when compared to abdominal USS, but is both less sensitive and specific than MRCP.[7,9] As it often affords better views of the distal CBD than USS,[10] and is more accessible than MRCP,[2] CT scanning plays an important role in the evaluation of patients with acute biliary dysfunction in settings where access to MRCP is limited.

Comparisons to pre-existing models

The models presented in this paper narrowly outperform the majority of other comparable models identified via a literature review[11–15] other than the ModelArts ExeML model by Zhang *et al*.[15] which reports an accuracy, sensitivity and specificity of 97%. However, reporting of the specific mechanisms and techniques used to achieve this accuracy is limited.

Limited sample size

Ongoing application of the models to larger sample sizes will ensure that the models are generalisable and will help to improve their accuracy. The models presented by this study boast highly accurate results despite the limited sample size, and the application of these models to larger and prospective datasets will offer an opportunity to improve this level of accuracy further. The inclusion of more diverse patient populations and healthcare settings will improve the models' validity across a wider range of clinical scenarios. Additionally, the use of larger sample sizes may allow the models to generate more nuanced outcomes via the identification of subgroups within the dataset who exhibit varying degrees of risk for choledocholithiasis.

Given patients with prior hepatobiliary surgery or known hepatobiliary disease or malignancy were excluded from the training data, it is likely that the models' generalisability to these populations is poor. However, this population likely represents a group of patients wherein the use of MRCP to evaluate acute-on-chronic presentations of biliary dysfunction is reasonable; the possible association between their acute presentation and their longstanding or historic hepatobiliary disease diminishes the probability of choledocholithiasis as the cause for their presentation and thus places them in an intermediate risk category, whereby MRCP is recommended as the choice of investigation.[4] As such, predicting the risk of choledocholithiasis in this patient population is beyond the scope of the models proposed by this study.

Clinical implementation and practical utility

Currently, the best performing model requires the input of 29 predictor variables. The integration of machine learning models into electronic health records (EHRs) has proven to be a viable approach in other medical contexts, enabling the generation of machine learning insights at the point of care; such models are designed to automatically populate predictor variables with data sourced from the EHR.[16,17] However, in their current state, the models are likely too cumbersome to be applied prospectively to incident cases in settings where values must be entered manually. Ongoing refinement and feature selection is required to develop a model that performs as well as the models presented here whilst utilizing fewer predictor variables. This can be achieved via ongoing application of the models to larger prospective datasets.

It is noted that models that generate a binary output regarding the presence of choledocholithiasis are unlikely to be especially clinically useful and as such ongoing development of more practical and informative outputs is warranted. This may include the development of a multi-category classification model. Such a model may classify a patient based on whether they are most likely going to benefit from having a laparoscopic cholecystectomy and intraoperative cholangiogram, an MRCP, or proceeding directly to ERCP.

Considerations regarding variable selection and importance

Although it would be an important finding clinically, the presence of malignancy on imaging is considered an unimportant variable by the RF models. This likely stems from the small number of patients with a malignancy that were included in the data ($n = 13$), as a variable's importance is determined by the change in a model's accuracy that is brought about by randomizing that variable across the dataset whilst keeping all other variables constant. It is assumed that this variable's importance will increase with models trained on a larger dataset featuring more patients with imaging evidence of malignancy.

High lipase levels were consistently negatively correlated with a final diagnosis of choledocholithiasis across all models, which may represent the speculated pathophysiology of gallstone pancreatitis; the passing of a gallstone through the ampulla of Vater causes the ampulla to be momentarily patulous, which begets the activation of pancreatic enzymes through the reflux of bile and intestinal enzymes and subsequent pancreatitis.[18] Whilst this may represent the models' ability to capture a complex relationship in the data, it should be noted that the models may underpredict choledocholithiasis in patients with concurrent pancreatitis.

A major limitation of this study was that, aside from bilirubin, all variables related to clinical findings and biochemical investigations were collected from the first recorded entries of a patient's admission. This means that the models were limited in their ability to capture trends in patients' admissions, instead capturing a cross-sectional representation of the patients at their initial presentation. Ongoing data collection and refinement of the models will facilitate the production of a novel model that considers longitudinal aspects of a patient's admission, allowing for trends in certain variables to influence the output of the models. Ultimately, this could lead

© 2024 Royal Australasian College of Surgeons.

14452197, 2024, 7-8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ans.18950 by University Of Southampton, Wiley Online Library on [18/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

to the development of an adaptive risk calculator that is applied on a day-by-day basis during a patient's admission, providing an evolving risk stratification as the patient's clinical picture changes.

Conclusion

The results of these machine learning models offer significant promise in the realm of diagnostic and management decision-making for patients with choledocholithiasis. Through the application of machine learning techniques, we have demonstrated that machine learning models exhibit exceptional performance in their effort to predict choledocholithiasis in patients presenting with acute biliary dysfunction, to the extent that their accuracy surpasses that of the ASGE choledocholithiasis risk stratification guidelines.

Although applied to relatively small datasets in this study, the clinical utility of these models is poised to grow through ongoing validation using larger datasets as well as through application to prospective patients. This utility will further improve as the models are refined to yield more practical outputs.

The ongoing enhancement and validation of these models will serve to further establish their credibility and applicability in optimizing patient care and decision-making processes in the management of acute presentations of choledocholithiasis and other biliary tract diseases.

Conflicts of interest

None declared.

Author contributions

Joshua Blum: Conceptualization; data curation; formal analysis; methodology; software; writing – original draft; writing – review and editing. **Sam Hunn:** Conceptualization; data curation; writing – review and editing. **Jules Smith:** Conceptualization; data curation; writing – review and editing. **Fa Yu Chan:** Data curation; writing – review and editing. **Richard Turner:** Conceptualization; supervision; writing – review and editing.

References

1. Sagvand BTea. Unnecessary MRCP prior to ERCP in patients with choledocholithiasis: the role of on-site ERCP. *Am. J. Gastroenterol.* 2020; **115**: 493.
2. Australia Co. Availability and accessibility of diagnostic imaging equipment around Australia. 2018.
3. Anand G, Patel YA, Yeh HC *et al*. Factors and outcomes associated with MRCP use prior to ERCP in patients at high risk for choledocholithiasis. *Can. J. Gastroenterol. Hepatol.* 2016; **2016**: 5132052.
4. Buxbaum JL, Fehmi SM, Sultan S *et al*. ASGE guideline on the role of endoscopy in the evaluation and management of choledocholithiasis. *Gastrointest. Endosc.* 2019; **89**: 1075–105.
5. Singhvi G, Ampara R, Baum J, Gumaste V. ASGE guidelines result in cost-saving in the management of choledocholithiasis. *Ann. Gastroenterol.* 2016; **29**: 85–90.
6. Rashtak S, Goyal H, Chandran A *et al*. Practice patterns and adherence to society guidelines for suspected choledocholithiasis: a comparison of academic and community settings from a large US healthcare system. *Front. Med. (Lausanne)* 2022; **9**: 1000368.
7. Woo LH *et al*. Diagnostic performance of the current risk-stratified approach with computed tomography for suspected choledocholithiasis and its options when negative finding. *Hepatobiliary Pancreat. Dis. Int.* 2019; **18**: 366–72.
8. Hunn S, Blum J, Smith J, Chan FY, Turner R, 2023, Patterns of MRCP utilisation in the diagnosis and management of choledocholithiasis: a quantitative retrospective assessment at a regional centre (unpublished local audit data). Royal Hobart Hospital, 2023.
9. You MG, Jung YY, Shin JY. Role of magnetic resonance cholangiopancreatography in evaluation of choledocholithiasis in patients with suspected cholecystitis. *J. Korean Soc. Radiol.* 2018; **78**: 147–56.
10. You MG *et al*. Ultrasonography of the distal common bile duct: an easy-to-perform technique for better visualization. *Arab J. Gastroenterol.* 2021; **22**: 203–8.
11. Akshintala VS, Tang B, Kamal A *et al*. Risk estimation, machine learning based ERCP decision-making tool for suspected choledocholithiasis. *Gastrointest. Endosc.* 2019; **89**: AB246–7.
12. Azizian JM, Dalai C, Trieu H, Rajan A, Tabibian JH. Id: 3520519 Logistic regression and machine learning methods predict choledocholithiasis more accurately compared to current criteria. *Gastrointest. Endosc.* 2021; **93**: AB145–6.
13. Cohen RZ, Tian H, Sauer CG *et al*. Creation of a pediatric choledocholithiasis prediction model. *J. Pediatr. Gastroenterol. Nutr.* 2021; **73**: 636–41.
14. Dalai C, Azizian J, Trieu H *et al*. Machine learning models compared to existing criteria for noninvasive prediction of endoscopic retrograde cholangiopancreatography-confirmed choledocholithiasis. *Liver Res.* 2021; **5**: 224–31.
15. Zhang HX, Gao JP, Sun Z *et al*. Diagnostic accuracy of updated risk assessment criteria and development of novel computational prediction models for patients with suspected choledocholithiasis. *Surg. Endosc.* 2023; **10**: 7348–57.
16. Feng A. A machine learning pipeline for accurate COVID-19 health outcome prediction using longitudinal electronic health records. *AMIA Annu. Symp. Proc.* 2022; **2021**: 448–56.
17. Li Y, Mamouei M, Salimi-Khorshidi G *et al*. Hi-BEHRT: hierarchical transformer-based model for accurate prediction of clinical events using multimodal longitudinal electronic health records. *IEEE J. Biomed. Health Inform.* 2022; **27**: 99–1117.
18. Arendt T, Nizze H, Monig H, Kloehn S, Stuber E, Folsch UR. Biliary pancreatic reflux-induced acute pancreatitis – myth or possibility? *Eur. J. Gastroenterol. Hepatol.* 1999; **11**: 329–35.

© 2024 Royal Australasian College of Surgeons.

14452197, 2024, 7-8, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ans.18950 by University Of Southampton, Wiley Online Library on [18/11/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License